GENERAL CHEMISTRY I LAB MANUAL

Jessica Thorpe
Meredith College

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General Chemistry I Lab Manual

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Licensing

A detailed breakdown of this resource's licensing can be found in ***Back Matter/Detailed Licensing****.*

CHAPTER OVERVIEW

0: Safety

0.1: Chemicals to be used

IMPORTANT NOTICE

Reagents used in CHE 141 are listed below. Safety Data Sheets for these reagents are on file in the lab. **IF YOU HAVE A MEDICAL CONDITION INCLUDING CURRENT PREGNANCY (OR YOU BECOME PREGNANT) THAT COULD AFFECT YOUR FULL PARTICIPATION IN THIS COURSE, PLEASE IMMEDIATELY CONSULT YOUR PERSONAL PHYSICIAN WITH THIS INFORMATION TO DETERMINE HOW EXPOSURE TO THESE CHEMICALS MIGHT AFFECT YOUR CONDITION AND THE POTENTIAL FOR HARMFUL EXPOSURE.** Please see instructor as soon as possible to discuss possible classroom accommodations.

NAME	**CAS No.**
Acetic acid	64-19-7
Acetone	67-64-1
Ascorbic acid	50-81-7
Barium chloride	10361-37-2
Calcium chloride	10043-52-4
Cupric nitrate	19004-19-4
Ethanol	64-17-5
Hexanes	110-54-3
Hydrochloric acid	7647-01-0
Glycerol	56-81-5
Iodine	7553-56-2
Lithium chloride	7447-41-8
Magnesium	7439-95-4
Potassium bromide	7758-02-3
Potassium carbonate	584-08-7
Potassium chlorate	3811-04-9
Potassium chloride	7447-40-7
Potassium hydrogen carbonate	298-14-6
Potassium nitrate	7757-79-1
Potassium oxalate	6487-48-5
Potassium permanganate	7722-64-7
Potassium phosphate	7758-53-2
Potassium sulfate	7778-80-5
Silver nitrate	7761-88-8
Sodium bicarbonate	144-55-8
Sodium carbonate	497-19-8
Sodium chloride	7647-14-5

Sodium hydroxide	1310-73-2
Sodium metal	7440-23-5
Strontium chloride	10025-70-4
Zinc	7440-66-6

0.2: Emergency Procedures

Emergency Procedures

Immediately alert the instructor to ANY fire or accident.

Fire Alarm

In case of a fire alarm, turn off all burners, water, and electrical devices. Collect any valuables (calculators, purses, etc.) and exit the building by assigned routes. DO NOT RUN. A copy of the evacuation route for your laboratory is posted on the laboratory bulletin board. Be familiar with the exit routes. Go to your assigned assembly area outside the building and remain there until the instructor has determined that all students are accounted for. Do not return to the building until given permission by your instructor or a campus security officer.

Injury

Immediately notify the instructor of ANY injury. First aid supplies are available in the laboratory. Minor injuries or burns may be treated by the laboratory instructor. Students with more serious injuries will be escorted to the student health center or a local hospital emergency room.

Medical Emergency Room

All medical emergencies (life-threatening situations, chest pain, excessive bleeding, eye injuries, head injuries, ingestion or inhalation of toxic substance, seizures, serious allergic reaction, suspected fracture) should be reported immediately to Wake EMS by dialing 9-911 from a campus phone and then Campus Police contacted by dialing 8888 from a campus phone or (919)-760-8888 from any other phone.

Emergency Information

Meredith College guidelines for emergency situations can be found online at the link below. It is your responsibility to read this brochure and be familiar with the information.

https://www.meredith.edu/emergency-planning/emergency-procedures/

Emergency Phone Numbers

Campus Police 8888

Fire, Police, Rescue 9-911

Campus Health Services 8535

Environmental Emergency 800-424-8802

Campus Maintenance 8560 (to shut off gas, water, or electricity if necessary)

0.3: Safety Dos and Don'ts

SAFETY REGULATIONS FOR CHEMISTRY LABORATORIES

DO THE FOLLOWING whenever you are in the laboratory:

1. Wear approved eye protection at all times. If you get a chemical in your eye, wash the eye with flowing water from the eye wash station for at least 15 minutes. Forcibly hold the eye open for effective washing.

2. Dress appropriately for lab. Wear clothing that will protect your skin. Pants, skirts, or dresses extend to the ankle with no skin showing. Sleeveless shirts and bare midriffs are not permitted. Shoes must cover the entire foot. Sandals, open shoes, or crocs are not permitted. Confine long hair.

3. Place backpacks in an area where they will be out of the way and do not present a tripping hazard or block access to emergency equipment (safety showers, fire extinguishers) or exits.

4. In case of fire or accident, call the instructor at once. Know the location of the fire extinguishers, fire blanket, eyewash stations, safety shower, and first aid kit, and know how to use them. If your clothing catches fire, immediately DROP to the floor and ROLL over to smother the flames. Never use a fire extinguisher of any type on a person; instead use a fire blanket to smother the flames.

5. Immediately report any injury, no matter how minor, to the instructor.

6. If a fire alarm sounds, turn off any gas valve or water that is in use. Collect your valuables and calmly exit the building by the nearest exit. Meet in the assembly area designated by your instructor and remain there until the instructor has determined that everyone is out of the building.

7. Be extremely careful lighting or using a gas burner. Never leave a burner unattended.

8. Carefully read labels on reagent bottles twice before using. Using the wrong chemical may have serious consequences. Consider all chemicals to be hazardous. Protective gloves will be provided for those experiments that require them. Safety Data Sheets (SDS) are available for all chemicals.

9. Clean up any spilled materials immediately. Notify the instructor to obtain any special instructions for cleaning.

10. Avoid breathing any fumes. Use the snorkel hoods at the lab stations or the fume hoods for any experiments that will generate fumes.

11. Follow the procedures for disposal of chemical wastes as given in the laboratory manual or by the instructor.

12. If any chemical contacts your skin, immediately wash it off with plenty of running water. The safety shower should be used only if a large area of your body is exposed to a corrosive or toxic chemical. Immediately remove any contaminated clothing.

13. Clean up broken glassware immediately and dispose of the glass in a container marked “BROKEN GLASS ONLY.” NEVER use glassware that is broken, cracked, or chipped.

14. When you are finished in the laboratory, thoroughly clean your work area with a sponge. Check to be certain that the gas and water are turned off and that all of your equipment is put away. Wash your hands thoroughly before leaving the laboratory.

DO NOT DO THE FOLLOWING when you are in the laboratory:

1. Do not eat, drink, smoke, or engage in horseplay.

2. Do not work in a laboratory unless an instructor is present. Never perform any unauthorized experiment.

3. Do not insert your spatula or pipet directly into a reagent bottle because it may result in contamination.

4. Do not allow papers to accumulate in your work area; they are a fire hazard.

5. Do not point a test tube containing a reacting mixture toward yourself or another person.

6. Do not add water to concentrated acids because the mixture may become extremely hot. Always add the acid slowly to water with constant stirring.

7. Do not remove chemicals from the laboratory.

8. Do not allow visitors.

9. Do not use cell phones.

10. Do not place calculators or computers where chemicals might be spilled on them.

0.4: Hazard Communication

Communication of Chemical Hazards

The Hazard Communication Regulation of the Occupational Safety and Health Administration (OSHA) requires that anyone who might come in contact with chemicals be informed of potential hazards. Communication of chemical hazards is through labels and safety data sheets as described below.

The Fire Diamond

A quick visual representation of hazards is provided by the Health, Flammability, Instability (HFI) or “Fire” diamond developed by the National Fire Protection Association (NFPA). The diamond consists of four quadrants with numbers or special symbols representing the degree of certain hazards. The top three quadrants contain National Fire Protection Association (NFPA) codes (numbers from 0 to 4) indicating the degree of a particular risk.

4 – Extreme Risk

3 – Serious Risk

2 – Moderate Risk

1 – Slight Risk

0 – Minimal Risk

Sample of an HFI Diamond

Interpretation of HFI Diamond

Quadrant	Type of Hazard	Color	Example
left	health	blue	3 = severe health hazard
top	flammability	red	4 = extremely flammable
right	instability	yellow	2 = moderate reactivity
bottom	special	white	~~W~~ = use no water

Hazard Pictograms

OSHA required hazard pictograms on labels consist of a black hazard symbol on a white background framed within a red border in the shape of a square set on a point. You should be familiar with these symbols.

Health Hazard  • Carcinogen • Mutagenicity • Reproductive Toxicity • Respiratory Sensitizer • Target Organ Toxicity • Aspiration Toxicity	**Flame**  • Flammables • Pyrophorics • Self-Heating • Emits Flammable Gas • Self-Reactives • Organic Peroxides	**Exclamation Mark**  • Irritant (skin and eye) • Skin Sensitizer • Acute Toxicity • Narcotic Effects • Respiratory Tract Irritant • Hazardous to Ozone Layer (Non-Mandatory)
Corrosion  • Skin Corrosion/Burns • Eye Damage • Corrosive to Metals	**Exploding Bomb**  • Explosives • Self-Reactives • Organic Peroxides	**Gas Cylinder**  • Gases Under Pressure
Flame Over Circle  • Oxidizers	**Skull and Crossbones**  • Acute Toxicity (fatal or toxic)	**Environment**  Not Mandatory • Aquatic Toxicity

For a more complete discussion of hazard pictograms, visit the OSHA Web site. www.osha.gov

0.5: Safety Data Sheets

Safety Data Sheet (SDS)

Chemical manufacturers are required by law to provide a Safety Data Sheet (or SDS) for each chemical. Prior to 2015, this document was called a Material Safety Data Sheet or MSDS. Although your instructor or laboratory manual will always inform you of any hazards associated with the chemicals you will be using, you may view the SDS for any chemical you will use in the laboratory. You should be familiar with the location of the SDS file.

The format of a Safety Data Sheet is standardized to contain 16 sections, each providing a specific type of information.

Section	Section Heading	Information
1	Identification of the substance or mixture and of the supplier	
2	Hazards identification	
3	Composition/information on ingredients Substance/Mixture	The CAS (Chemical Abstract Service) number is a unique identifier and is especially important when accessing computer databases.
4	First aid measures	
5	Firefighting measures	
6	Accidental release measures	
7	Handling and storage	This section gives advice on proper storage of the chemical.
8	Exposure controls/personal protection	
9	Physical and chemical properties	
10	Stability and reactivity	These sections provide information about chemical incompatibilities and other chemical reaction dangers.
11	*Toxicological information*	Common abbreviations used in this section are listed below. Normally chemical exposure involves a chemical entering your body through inhalation (breathing), eye contact, skin absorption, or orally (ingestion). HMN: Human IVN: Intravenous (in the blood stream) MUS: Mouse ORL: Oral SKN: Skin TC_{Lo}: Toxic concentration low TD_{Lo}: Toxic dose low LC_{50}: Lethal Concentration 50; The concentration of a substance in air that can kill 50% of the test animals when administered as a single respiratory exposure LD_{50}: The lethal dose that will cause death in 50% of the test subjects.
12	Ecological information *(non mandatory)*	
13	*Disposal considerations (non mandatory)*	
14	*Transport information (non mandatory)*	
15	*Regulatory information (non mandatory)*	
16	Other information including information on preparation and revision of the SDS	

To search for a SDS for a chemical, go to http://www.ilpi.com/msds/index.html

0.6: SDS Activity

Using the sample SDS for sodium bicarbonate (baking soda), find the following information.

https://www.fishersci.com/content/dam/fishersci/en_US/documents/programs/education/regulatory-documents/sds/chemicals/chemicals-s/S25533.pdf

CAS Number ________________ Chemical Formula ________________

First Aid (Eye Contact) ____

Eye Protection ____

NFPA Ratings Health _____ Fire _____ Instability _____ Special _____

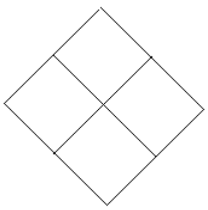
Signal Word ______________________________________

Hazard Statements ____

____

Hazard Pictogram ____

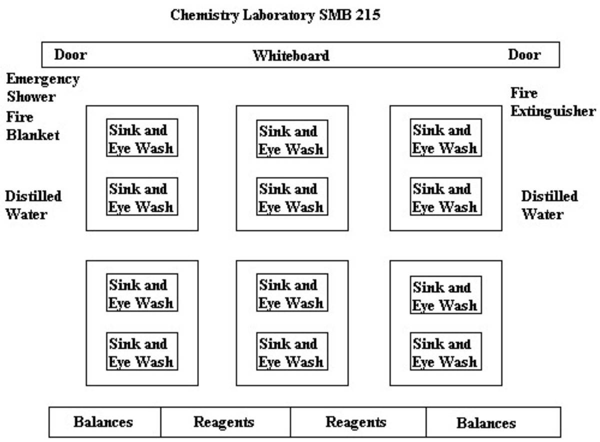
Complete the fire diamond for sodium bicarbonate.

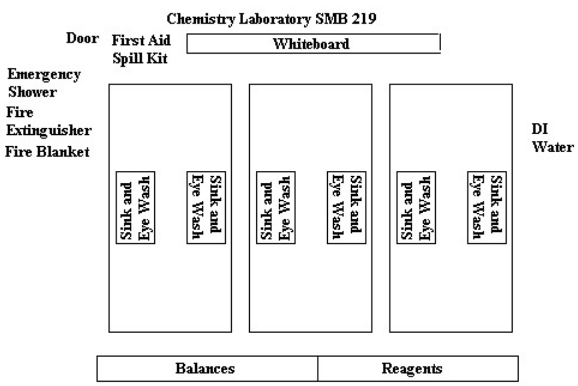


![LibreTexts logo]

0.7: Lab Layout

Laboratory Layout





0.8: Check in/out

Check in

Remove all the equipment from your lab drawer and cupboard. Go through the list of equipment and identify each piece. If the glassware is clean, return it to the drawer or to the cupboard.

If any glassware is chipped, cracked, or broken, then discard it in the container marked BROKEN GLASS ONLY.

If any of your glassware or equipment is missing or broken, write the name of the item on a piece of paper with your name. The instructor or a laboratory assistant will get the missing item for you.

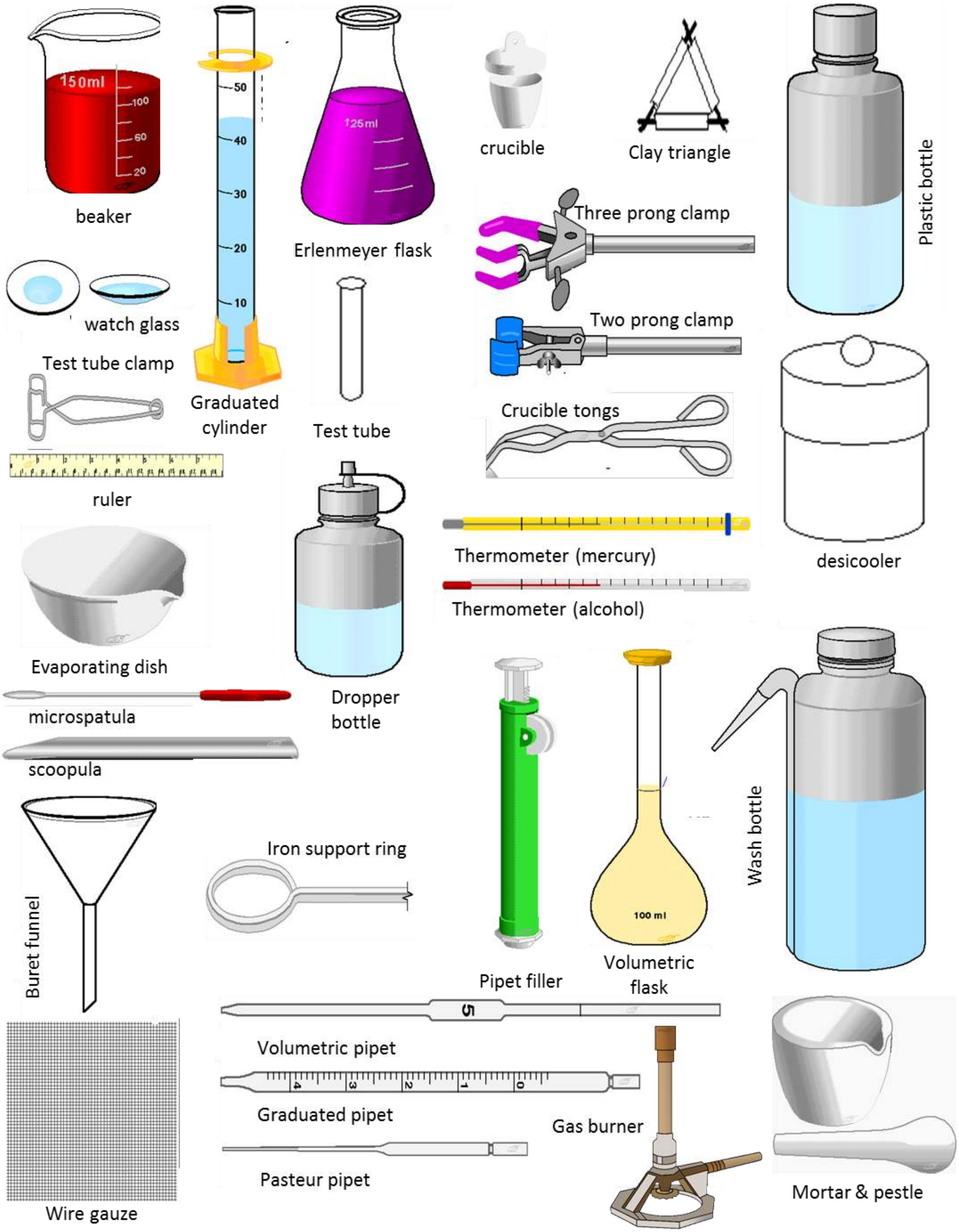
If you have extra items, place them in the tray designated for excess equipment.

If any piece of glassware is dirty, clean it with the soap solution in the spray bottle at your bench. Dry the outside with a paper towel, invert on a paper towel and allow most of the water inside to drain out before returning the glassware to the drawer or cupboard.

Quantity	**Item/Quantity**	Size/Description
Equipment in Drawer		
6	Beakers	2 – 50 mL 2 – 100 mL 1 – 150 mL 1 – 250 mL
1	Buret Card	
1	Evaporating Dish	
1	Flask, Erlenmeyer	50 mL
3	Flasks, Erlenmeyer	250 mL
1	Graduated Cylinder	10 mL
1	Graduated Cylinder	50 or 100 mL
2	Litmus paper	1 red, 1 blue
3	Pipets	Small
2	Rubber bulbs	
1	Ruler	
1	Spatula/Scoopula	
5	Stirring rods	4 small, 1 large
1	Test tube clamp	
1	Test tube rack	
17	Test Tubes	2 – 150 x 18 mm 3 – 100 x 13 mm 12 – 75 x 10 mm
1	Watch glass	
Community Equipment in Cupboard		
2	Bottles	Plastic, with lids – 1 L
2	Brushes	1 small, 1 large

1	Buret Funnel	
1	Buret holder	
1	Burner	with rubber tubing
1	Clay triangle	
1	Dessicooler	
1	Flask clamp	
2	Flasks, Volumetric	100 mL
1	Iron ring	
1	Thermometer	
1	Tongs	
1	Wash bottle	
2	Wire gauze	

0.9: Lab equipment



CHAPTER OVERVIEW

1: Experiment 1- Introduction to Chemical Reactions

Experiment 1: Introduction to Chemical Reactions

1.1: Background Information

Chemists summarize the results of chemical reactions by writing chemical equations. For example, carbon combines with oxygen to form a new substance called carbon dioxide. A chemist would summarize the results of this reaction by writing the following chemical equation.

$C(s) + O_2(g) \rightarrow CO_2(g) + \text{heat}$

The substances on the left of the arrow are called **reactants**, those on the right, **products**. Heat is given off in this reaction.

The symbols used for elements are simply those found in the periodic table, thus carbon is represented by C. Oxygen is represented by the symbol O_2 because we do not normally find single atoms of pure oxygen; oxygen atoms are usually found bound in units of two called **diatomic molecules**. Other common diatomic molecules are H_2, N_2, F_2, Cl_2, Br_2, and I_2. A compound is represented by writing the symbols for the elements of which it is composed side-by-side. For example, the compound carbon dioxide is written as CO_2. The subscript numbers represent the number of atoms of each element found in a molecule of the compound. Ones are not written but understood to be present. Thus in a molecule of CO_2, there are two atoms of oxygen and one atom of carbon.

There is more to writing a chemical equation than simply writing the chemical formulas for the reactants and products. The equation must obey the law of conservation of mass; atoms cannot be created nor destroyed. For example, the equation

$H_2(g) + O_2(g) \rightarrow H_2O(l)$

is not balanced. There are two oxygen atoms on the left and only one on the right. To balance chemical equations, numbers (**coefficients**) are placed in front of the symbol for each substance such that there will be equal numbers of atoms on both sides of the equation. The above equation may be balanced as follows

$2\ H_2(g) + O_2(g) \rightarrow 2\ H_2O(l)$

If a chemical formula does not have a number in front of it, such as O_2, then a one is understood to be present. The equation is now balanced with four hydrogen atoms and two oxygen atoms on each side.

The **state** of a substance is often represented by a symbol in parentheses next to the symbol for that substance. The symbols commonly seen are

(s) solid

(*l*) liquid

(g) gas

(aq) aqueous solution (the substance is dissolved in water)

Many chemical reactions can be classified into one of four types:

I. **Combination** Reactions - These reactions may be represented by the general equation A + B → AB. There are always multiple reactants and only one product. Some specific examples are

$Fe(s) + S(s) \rightarrow FeS(s)$

$C(s) + O_2(g) \rightarrow CO_2(g)$

II. **Decomposition** Reactions - A decomposition reaction may be represented by the general equation AB → A + B. This type of reaction always has one reactant but multiple products. Some examples are

$CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$

$PCl_5(g) \rightarrow PCl_3(g) + Cl_2(g)$

III. **Single Displacement** Reactions - A reaction of this type may be represented by the general equation A + BC → AC + B. In these reactions, a pure element reacts with a compound to form a different pure element and a compound. Some specific examples are

$Zn(s) + H_2SO_4\ (aq) \rightarrow ZnSO_4(aq) + H_2(g)$

$CuSO_4(aq) + Zn(s) \rightarrow ZnSO_4(aq) + Cu(s)$

IV. **Double Displacement** Reactions - A reaction of this type may be represented by the general equation AB + CD → AD + CB. These reactions are always two compounds reacting together to make two different compounds. Some examples are

$BaCl_2(aq) + Na_2SO_4(aq) \rightarrow BaSO_4(s) + 2NaCl(aq)$

$KOH(aq) + HNO_3(aq) \rightarrow KNO_3(aq) + H_2O(l)$

1.2: Safety

Chemicals	**Hazards**	**Pictograms**	**Required PPE**
Sodium Metal	Contact with water releases flammable gases Severe skin burns and eye damage		Glasses/Goggles Gloves
Potassium Chlorate	May cause fire/explosion Harmful if swallowed and inhaled	 	Glasses/Goggles Gloves
Magnesium Strips	Catches fire spontaneously		Glasses/Goggles Gloves

1.3: Procedure

CAUTION: The following reaction will be conducted as a demonstration by your instructor. Do not attempt to do this reaction yourself.

a) Put potassium chlorate ($KClO_3$) in a 100 x 13 mm test tube to a depth of one centimeter and heat strongly by holding the tube at approximately a 45^o angle in a Bunsen burner flame. The test tube should be slowly rotated so that the contents are heated evenly. As soon as bubbles begin to evolve from the molten material, put a glowing splint at the mouth of the tube. Based on the elements in the reactants, the evolving gas might be H_2, O_2 or CO_2. Hydrogen, oxygen, and carbon dioxide are colorless, odorless gases. Hydrogen is flammable, and a burning wooden splint will ignite the hydrogen (sometimes with a small pop (explosion). Oxygen supports combustion, and a glowing wooden splint will burst into flames when place in pure oxygen. Carbon dioxide does not support combustion and will extinguish a glowing wooden splint. Many fire extinguishers use carbon dioxide.

The balanced chemical equation is:

$2\ KClO_3(s) + heat \rightarrow 2\ KCl(s) + 3\ O_2(g)$

Classify this reaction: ________________________________

Describe the behavior of the glowing splint, and indicate what this tells us.

Label each substance as an element or a compound and specify its state (solid, liquid, gas, or aqueous):

$KClO_3(s)$:

$KCl(s)$:

$O_2(g)$:

CAUTION: The following reaction will be conducted as a demonstration by your instructor. Do not attempt to do this reaction yourself.

b) Cut a piece of sodium metal the size of a match head. Handle the sodium with forceps and drop it in a test tube containing several milliliters of water. Point the test tube away from you, toward the back of a fume hood. Note the immediate violent reaction that occurs when the sodium strikes the water. Hold a <u>burning</u> wooden split over the mouth of the test tube. The balanced chemical equation is:

$2\ Na(s) + 2\ H_2O(l) \rightarrow 2\ NaOH(aq) + H_2(g) + heat$

Classify this reaction: ________________________________

Describe the behavior of the burning wooden splint, and indicate what this tells us.

Label each substance as an element or a compound and specify its state (solid, liquid, gas, or aqueous):

$Na(s)$:

$H_2O(l)$:

$NaOH(aq)$:

$H_2(g)$:

CAUTION: Do not look directly at the magnesium while it is burning. The intense light could hurt your eyes.

c) Take a piece of magnesium ribbon, tongs, and a watch glass to the fume hood with a Bunsen burner. Hold the piece of magnesium ribbon with tongs and set fire to it with the nonluminous flame of the burner. Have a watch glass readily available. When the magnesium begins to burn, hold it over the watch glass and collect the ash. The white powder formed is mostly magnesium oxide, although a small amount of magnesium nitride is also formed. The magnesium reacted with the oxygen and nitrogen in the air. The balanced chemical equations are:

$2\ Mg(s) + O_2(g) \rightarrow 2\ MgO(s) + light$

$3\ Mg(s) + N_2(g) \rightarrow Mg_3N_2(s) + light$

Classify these reactions: ________________________________

Label each substance as an element or a compound and specify its state (solid, liquid, gas, or aqueous):

$Mg(s)$:

$O_2(g)$:

$MgO(s)$:

$N_2(g)$:

$Mg_3N_2(s)$:

CHAPTER OVERVIEW

2: Experiment 2- Science of Measurements

2.1: Background Information

The scientific method reduced to its simplest form is learning by experience. We experience the physical world through our five senses. In science these sensations of sight, sound, touch, taste and smell are classified as observations. Observations are the first step in the scientific method. Any conclusions that we reach are no better than the care we use in making our observations. If we do not exercise care in making observations, then any conclusions that we reach may be incorrect; therefore, it is imperative to always make observations with as much care as possible.

Observations that do not require the use of numbers are said to be qualitative. For example, "It was very warm this morning" is a qualitative observation. Observations that require numbers are said to be quantitative. "It was 80 degrees this morning" is a quantitative observation. When our observations become quantitative, we say that we are making measurements. Since chemistry is an empirical (experimental) science, most of the experiments will involve measurements.

Units of Measurement

All interval and ratio scale measurements include a number indicating the magnitude of the measurement and a unit that indicates the physical quantity (length, mass, time, etc.) being measured. For example, the measurement 2.6 has no meaning by itself. However, if we add the unit “inches” to the 2.6, then we know that we have measured some length or distance. A measurement must always include the appropriate unit. There are several systems of units. In science we use the metric system known as the International System of Units (abbreviated SI). The seven base units for the SI system and their symbols are listed below.

Physical Quantity	Unit Name	Unit Symbol
length	meter	m
mass	kilogram	kg
time	second	s
temperature	kelvin	K
amount of substance	mole	mol

Scientific Notation

Many numbers used in science are extremely large or extremely small. For example, the distance from the earth to the sun is about 150,000,000,000 meters, and the size of a lithium atom is approximately 0.000000000268 meters. Large and small numbers such as these are more conveniently expressed using scientific notation. Scientific notation is a type of exponential notation in which only one digit is kept to the left of the decimal point. For example, the earth-sun distance can be written as:

1.5 x 100,000,000,000 m or 1.5 x 10 x 10 x 10 x 10 x 10 x 10 x 10 x 10 x 10 x 10 x 10 m

or, in scientific notation, 1.5 x 10^{11} m.

The exponent **11** indicates the number of places that the decimal place has been moved to the left. Similarly, the size of a lithium atom may be written as

2.68 m / 10,000,000,000 or 2.68 m /(10 x 10 x10 x 10 x 10 x 10 x 10 x 10 x 10 x 10)

or, in scientific notation, 2.68 x 10^{-10} m.

Metric Prefixes

The metric system uses a set of prefixes to represent quantities that are larger or smaller than the standard base unit. Some of the more commonly used metric prefixes are listed in the table below.

Prefix	Symbol	Factor	Power of Ten
kilo	k	1,000	10^3
deci	d	0.1	10^{-1}
centi	c	0.01	10^{-2}

milli	m	0.001	10^{-3}
micro	μ	0.000001	10^{-6}
nano	n	0.000000001	10^{-9}

A centimeter is one hundredth of a meter.

1 cm = m = 0.01 m or 100 cm = 1 m

A millimeter is one thousandth of a meter.

1 mm = m = 0.001 m or 1,000 mm = 1 m

A kilogram is one thousand grams.

1 kg = 1,000 g or kg = 0.001 kg = 1 g

Reading Scales and Uncertainty in Measurement

Any experimental measurement will have some error associated with it. Here the word "error" does not mean "mistake" but simply the uncertainty that cannot be avoided in making a measurement. We never can be certain that we have determined the "true value" of a quantity.

When making measurements, it is important to carefully examine the scale being used and do two things:

1. Determine the direction to read the scale. Does the scale read from left to right or right to left? Does the scale read from top to bottom or bottom to top? This may seem like a trivial matter, but it is often a source of error.

2. Determine the smallest division on the scale. The smallest scale division is known as the **least count**. Always read the scale to one more decimal place than the least count. For example, the least count (smallest division) on the scale shown below is 1 cm; therefore, we can make a measurement with the scale to the nearest 0.1 cm.

|____|____|__l_|____|____|

0 1 2 3 4 5 cm

If the red line represents the end of an object being measured, you would have to estimate between the 2 cm and 3 cm divisions. The simplest way to do this is to imagine 10 smaller divisions between the 2 and 3 cm marks and then estimate the next digit in the measurement. 2.7 cm would be a reasonable estimate for this measurement. Note that the measurement has one more decimal place than the least count, and that the last place is an estimate and, therefore, there is some uncertainty associated with it.

The smaller the scale divisions (least count) the more precise the measurement that can be made. For example, the metric ruler shown above is less precise than the one you have in your laboratory desk drawer, since your laboratory ruler can be used to make measurements to the nearest 0.01 cm, while the one shown above can be used to make measurements to only 0.1 cm.

Significant Figures

All measurements have some uncertainty; therefore, any results obtained from computations using these measurements must be uncertain. There are statistical methods for determining this uncertainty; however, we often estimate these uncertainties using the method of significant figures. Significant figures indicate the "exactness" with which a measurement has been made. When you make a measurement, the digits you obtain directly from the measuring instrument, followed by the first estimated (or uncertain) digit of the measurement, are called the significant figures. The use of significant figures to estimate the uncertainty of the results of computations is useful when a more extensive analysis of uncertainty is not necessary.

Examples of some measurements and the number of significant figures for each measurement are given in the table below. To determine the number of significant figures in a measurement, read the number from left to right and count all digits starting with the first *non-zero* digit. Do not count any exponential part.

<u>Measuring Device</u> <u>Least Count</u> <u>Typical Measurement</u> <u>Significant Figures</u>

Ruler 0.1 cm 8.07 cm 3

15.36 cm 4

Buret 0.1 mL 6.42 mL 3

40.40 mL 4

100 mL Graduated Cylinder 1 mL 15.3 mL 3

60.0 mL 3

Thermometer 1 ^oC 3.1 ^oC 2

103.5 ^oC 4

Some other examples of measurements and significant figures are given below.

0.0023 g = 2.3 x 10^{-3} g (2 significant figures)

360.4 mL = 3.604 x 10^2 mL (4 significant figures)

1200 mg = 1.2 x 10^3 mg (2 significant figures)

The trailing zeroes in a non-decimal number such as 1200 are assumed **not** to be significant unless there is specific information to indicate that they are. If you mean for all the figures in this measurement to be significant, then it should be written with a final decimal, such as 1200. or written in scientific notation, 1.200 x 10^3.

Always remember that the concept of significant figures is related to the uncertainties in measured quantities and applies **only** to measured quantities. Counting numbers or defined quantities are exact. By definition there are exactly 12 eggs in a dozen, and there are exactly 100 cm in a meter. Since these quantities are exact, they can be treated as if they have an infinite number of significant figures. There is no uncertainty in them.

Measurements and Statistics

No measurement is exact. All measurements have error associated with them. The word "error" in the sense that it is used with measurements does not mean mistake, but the uncertainty associated with a measurement. One way that we often attempt to minimize the error associated with a measurement is to repeat the measurement a number of times and determine an average value or the mean. The mean value x_m is computed as follows:

$$x_m = \Sigma x_i / n$$

where x_i represents an individual measurement and n is the number of measurements.

The accuracy of a result is a measure of the agreement of a measurement or experimental result with the true value. Since the true value of a quantity can never be known exactly, an accepted value is commonly used for comparison. In experimental work the mean is an estimate of the true value. The absolute error associated with a result is the difference between the result, x_m, and the true or accepted value, μ.

absolute error = $x_m - \mu$

The relative error is the absolute error divided by the true value. The is often expressed as a percent and is known as the percent error.

percent error = $[(x_m - \mu) / \mu] \times 100\%$

Precision is a measure of the agreement between two or more measurements. Measurements or experimental results that are very nearly the same are said to be reproducible and are precise. One of the characteristics of good experimental work is that it is reproducible. We can begin to get a measure of precision by looking at how much each measurement differs from the mean. A quantity known as the deviation, d_i, is the difference between a member of a data set, x_i, and the mean value of the data set.

$d_i = x_i - x_m$

A quantity called the sample standard deviation, s, is a common statistical measure of precision.

$s = [\surd\ \Sigma(x_i - x_m)^2] / n - 1$

For small sets of data (n < 20), the standard deviation can be estimated from the range. This means of estimating the standard deviation is nearly as reliable as the value calculated using more rigorous statistical methods. The range is the difference between the largest value, x_L, and the smallest value, x_S.

range = $x_L - x_S$

For small data sets, the standard deviation, s, can be estimated using the following equation.

$s = \text{range} / \sqrt{n}$

For some experiments you will be graded on both your accuracy (how close your experimental value is to the correct value), and your precision (how close your repeated values are to each other). Science requires that you make observations and measurements with as much care as you possibly can.

SAMPLE STATISTICAL CALCULATION

Consider the results of an experiment for the triplicate analysis of the sulfur content of a sample. The results are summarized below.

experiment	measurement	deviation	$(\text{deviation})^2$
1	50.70 %	-0.01 %	0.0001
2	50.65 %	-0.06 %	0.0036
3	50.77 %	+0.06 %	0.0036
	x_m = 50.71 %		$S(\text{deviations})^2 = 0.0073$

Using the statistical formula for sample standard deviation gives

s = = ± 0.06 %

Using the range formula for estimating the sample standard deviation gives

s = = ± 0.07 %

Note that the two methods for estimating the standard deviation are very close. Therefore the mean is 50.71 % sulfur with a standard deviation of ± 0.07 %.

To perform statistical calculations, you can use the features of most scientific calculators or the statistical functions of spreadsheet software such as Microsoft Excel.

2.2: Pre-lab

Prior to coming to the laboratory session, carefully read The Science of Measurements and complete the following exercise.

1. Go to the Web site http://science.widener.edu/svb/tutorial/scinot.html and do enough exercises on scientific notation to feel confident with the concept.

There will be no pre-lab quiz for this experiment. The Science of Measurements experiment will not be collected for a grade. Please use this experiment to study for the Science of Measurements quiz.

2.3: Safety

Hazards	Pictograms	Required PPE
Glassware	n/a	Glasses/Goggles

2.4: Procedure

Scientific Notation Exercises

Write each of the following numbers using scientific notation.

1. 823,000 = _______________

2. 0.000000000818 = _______________

Write each of the following numbers without exponents.

1. 3.88247 x 10^4 = _______________

2. 5.0070 x 10^{-2} = _______________

Measurement Scales and Least Count

Examine each of the following measurement devices in your desk and record the least count and the number of decimal places to which the device's scale can be read.

Measuring Device	**Least Count** (smallest scale division)	Number of Decimal Places To Be Read
Plastic Ruler		
Small Graduated Cylinder		
Large Graduated Cylinder		
Thermometer		

Your instructor will assign a metal cylinder to your group. Each person in the group should independently measure the length and diameter of the cylinder. Use *centimeters* as the unit of length for these measurements. Record your results below. Remember that you can read the ruler to one more decimal place than the least count you determined in the previous exercise. Include the proper *units* with your measurements.

type of metal ____________________

length = __________________ diameter = ____________________

After all members of the group have made their measurements, get together and compare your measurement. Do all of you have exactly the same measurements? Discuss the reasons for any discrepancies among the members of your group. If anyone has a measurement that is significantly different from the rest of the group, that student should repeat any suspect measurement.

Significant Figure Exercise

How many significant figures are in your measurements of the length and diameter of the cylinder?

number of significant figures in length: _____

number of significant figures in diameter: ______

Conversion Exercises:

Convert the length measurement of the cylinder to meters. Show your calculations. Record your answer in the blank.

____

Convert the diameter measurement of the cylinder to meters. Show your calculations. Record your answer in the blank.

____

Convert the length measurement of the cylinder to millimeters. Show your calculations. Record your answer in the blank.

____

Convert the diameter measurement of the cylinder to millimeters. Show your calculations.

____

Determination of Volume using ruler measurements

You can determine the volume of your sample from your measurements of length and diameter. The volume of a cylinder is the area of the base multiplied by the length. The base of a cylinder is a circle with a radius, r; therefore, the area is πr^2. Since the diameter, d, was measured and r = d/2, then the volume of the cylinder is

$V = (\text{area of base}) \times (\text{length}) = (\pi r^2)\ell = \pi(d/2)^2\ell$ or $V = \pi(d^2/4)\ell$

where ℓ is the length of the cylinder.

Calculate the volume of the cylinder using your measurements of length and diameter in centimeters. Show the volume equation with the measurements you used for the variables. Include units in your calculations and express the volume with the appropriate units.

Volume = __________

What is the volume of your cylinder in milliliters? Show your calculations. Record your final answer in the blank.

Volume = __________

What is the volume of your cylinder in liters? Show your calculations. Record your final answer in the blank.

Volume = __________

Determination of Volume using water displacement

Fill your large graduated cylinder about half full with water. Read the level of water in the graduated cylinder. Remember to read the bottom of the meniscus. Record your reading. Now tilt the graduated cylinder sideways and allow your metal cylinder to slide slowly down the side of the graduated cylinder into the water. Place the graduated cylinder upright on the lab bench. Make sure there are no air bubbles clinging to the metal cylinder or graduated cylinder. If there are, use a glass stirring rod to try to remove them. Allow any water on the side of the graduated cylinder to completely drain down. Now read the level of water in the graduated cylinder again. Record the reading. The difference between the two reading is the volume of the metal cylinder.

Final volume reading with metal cylinder __________

Initial volume reading without metal cylinder __________

Volume of metal cylinder __________

Remove the metal cylinder and dry it completely. Pour the water from the graduated cylinder. Compare the volume obtained using the ruler measurements with the volume obtained by water displacement. Which volume do you think is more accurate? Why?

Determination of Mass

The top loading balances in our laboratory measure mass to three decimal places. Laboratory balances typically determine masses in grams. Instructions on how to use the laboratory balance are provided in the Laboratory Handbook. As a group, determine the mass of your metal cylinder and record the result below.

mass = _______________

Express the mass in milligrams. Show your calculations. Record your final answer in the blank.

mass = __________

Express the mass in kilograms. Show your calculations. Record your final answer in the blank.

mass = __________

Density

Density is one of the physical properties that can be used to characterize a substance. The densities of many substances can be found in chemical reference books, such as the *CRC Handbook of Chemistry and Physics*. Density, represented by the symbol, d, is defined as the mass m per unit volume, V.

d = m / V

Density is an intensive property, that is, it does not depend on the amount of matter. The density of a small cylinder has the same as the density of a large cylinder of the same material.

Calculate the density of your metal cylinder in units of grams per milliliter. Use the volume obtained from the ruler measurements. Show your calculations.

density = __________

List density value for each member of your group and calculate the average density for your group. Show your calculations.

average density = x_m = __________

Do you think that the average value you obtained for the density of your metal is the actual, exact density for this metal? Why or why not? Discuss these questions with the members of your group and list some of your answers.

Statistical Calculations

Use a handbook to look up the accepted value for the density of your metal cylinder.

Handbook value for density = __________

Reference:____

Calculate the absolute error for your experimental value for the density. Show your calculations.

absolute error = __________

Calculate the percent error for your experimental value for the density. Show your calculations.

% error = __________

Based on your % error, would you consider your density value to be accurate? Explain.

Calculate the range for your group's set of density values. Show your calculations.

range = __________

<u>Using the range</u>, estimate the sample standard deviation for your group's density data.

estimated standard deviation = s = __________

Based on your group's standard deviation, do you consider your group's data to be precise? Explain.

CHAPTER OVERVIEW

3: Experiment 3- Atomic Structure and Isotope Abundance

3.1: Background Information

All atoms have a positively charged core, the *nucleus*, with one or more negatively charged electrons moving about it. The electrical attractive force keeps the extremely fast moving electrons from escaping from the atom. The nucleus is composed of two kinds of particles collectively known as *nucleons*. *Protons*, which have a positive charge, and *neutrons*, which have a no charge, are the two kinds of nucleons. The positive protons exert strong repulsive forces on each other; however, the strong nuclear force holds the nucleons together to form the nucleus. Since atoms are electrically neutral, they have the same number of electrons as protons. The masses of subatomic particles are often represented in *atomic mass units* and the charges are represented as multiples of the charge on the electron (see Table 1). The nucleus accounts for most of the mass of an atom.

atomic mass unit = u = 1.66054 x 10^{-27} kg

elementary charge = e = 1.602177 x 10^{-19} C

Table 1: Properties of Subatomic Particles

particle	**symbol**	**unit charge /e**	**mass /u**
proton	or p	+1	1.00728
neutron	or n	0	1.00866
electron	or e	-1	0.000548680

The *atomic number* is the number of protons in a nucleus and is represented by the symbol Z. The *mass number* is the total number of nucleons (protons plus neutrons) and is represented by the symbol A. In general, an atom of an element can be represented using the following notation.

$$^{A}_{Z}\textbf{element symbol}$$

A specific example is ^{12}C, which represents a carbon atom with 6 protons and 6 neutrons or a total of 12 nucleons. This atom also has 6 electrons. For a particular element, the number of neutrons in the nucleus can vary. Atoms of the same element with different mass numbers are called *isotopes*. For example, there are three naturally occurring isotopes of potassium (see Table 2).

Table 2: Natural Isotopes of Potassium

Symbol	**protons**	**neutrons**	**electrons**	**% abundance**
	19	20	19	93.2581
	19	21	19	0.0177
	19	22	19	6.7302

3.2: Pre-lab

1. Atomic structure: Complete the following table:

Table 3: Atomic Structure Pre-Laboratory Practice

Protons	Neutrons	Electrons	Atomic Number	Mass Number	Atomic Symbol
6	**7**				
		42		**96**	
			55	**133**	

2. An element consists of the following isotopes:

Individual Isotope Mass Abundance

31.97207 amu 95.02%

32.97146 amu 0.75%

33.96786 amu 4.21%

35.96709 amu 0.02%

Calculate the weighted average atomic mass and identify the element.

3.3: Safety

There are no hazards in this experiment.

3.4: Procedure

Part 1: Scientific Data Analysis and Identification

1. Using Microsoft Excel, create a plot graph based on the first ten elements of the periodic table (Hydrogen through Neon) using atomic mass as the y-axis and atomic number as the x-axis.

2. How to make a graph in Excel:

a. Open Excel on your laptop or lab computer. Choose a blank workbook to open.

b. In column A, enter the atomic number of each element beginning with Hydrogen and continuing through Neon (1-10). This is the independent variable.

c. In column B, enter the atomic mass from the periodic table for each atomic number. This is the dependent variable.

d. Select both columns and click on Insert: Chart: XY Scatter.

e. Click anywhere on the chart and three small boxes will appear on the right. Click on the top box with a plus sign and select "Trend line".

3. Have your professor or TA check your graph before proceeding to the next step and write the trend line in the space provided here.

Trend Line: ____

4. Obtain and weigh an element model labeled for Part 1 and record it in the space given. Do not move the set away from the balance area.

Set # ______________ Model Mass (g) of unknown element: ____________________

5. Convert the model's mass in grams to individual isotope mass in atomic mass units (amu) using a 1:1 ratio and record the mass in Table 4.

6. Calculate the atomic number of the element using the mass of the element and trend line from the Excel graph and record it in Table 4. Show all work in the space provided.

7. Determine the unknown element and record its chemical symbol in Table 4.

8. Fill in the number of protons, neutrons, and electrons for the element in Table 4.

Table 4: Isotope Trend Line Results

Isotope Mass (amu)	Atomic Number	Chemical Symbol	# of protons	# of neutrons	# of electrons

Lab Procedure Part 2A: Identification Using the Periodic Table

1. Obtain a model set labeled for Part 2A by the balance. Do not move it away from the balance area. Weigh each isotope and record the mass and color of each in Table 5.

Set #: ________________

2. Locate the card with the relative abundance in the model set container. Record the value in Table 5.

3. Convert the model's mass in grams to isotope mass in atomic mass units (amu) using a 1:1 ratio and record each mass in Table 5.

4. Calculate the weighted average atomic mass of your unknown element given the mass of each isotope and its abundance from Table 5. Show your work in the space provided below. Record the atomic mass in Table 6. Locate the element on the periodic table and record its chemical symbol in the table.

Table 6: Unknown Element Identification

Atomic Mass (amu)	Element Name	Chemical Symbol

5. For each isotope, find the number of protons, neutrons and electrons and write it in Table 5.

Table 5: Model Isotopes

Color of Isotope Model	Model Mass (g)	Isotope Mass (amu)	Relative Abundance	# protons	# neutrons	# electrons

Lab Procedure Part 2b: Calculation of Abundance

1. Obtain a model set labeled for Part 2B by the balance. Do not move it away from the balance area. Record the set # and element information.

Set #: ________________ Element Name: ________________

Chemical Symbol: ________________

2. Record the atomic mass for the element **from the periodic table**.

Weighted Average Atomic Mass: ________________

3. Weigh each isotope and record the mass and color of each isotope in Table 7.

4. Convert the model's mass in grams to isotope mass in atomic mass units (amu) using a 1:1 ratio and record each mass in Table 7.

5. Calculate the abundance of each isotope by using the mass of each isotope and the average atomic mass. Show all work. Record the relative abundances in Table 7.

6. For each isotope, find the number of protons, neutrons and electrons and write it in Table 7.

Table 7: Percent Abundance Calculations

Color of Isotope Model	Model Mass (g)	Isotope Mass (amu)	Relative Abundance	# protons	# neutrons	# electrons

CHAPTER OVERVIEW

4: Experiment 4- Atomic Emission Spectroscopy

4.1: Background Information

The electromagnetic spectrum below shows the various kinds of radiant energy in order of decreasing energy.

Table 1: Electromagnetic Spectrum

gamma	x-ray	ultraviolet	visible	infrared	microwave	radio

high energy low energy

The radiant energy emitted or absorbed by substances has provided chemists with detailed information about the structure of matter. The study of the radiant energy emitted or absorbed by matter is known as **spectroscopy**. When energy is **absorbed** by matter, the atoms or molecules are promoted to higher energy states. When energy is **emitted** by matter, the atoms or molecules are left in lower energy states. For example, when a hydrogen atom absorbs energy, an electron is promoted from a lower energy state to a higher energy state. Since the electron has a greater kinetic energy, it is able to move farther from the nucleus and the atom is larger. When a hydrogen atom emits energy, the electron drops from a higher energy state to a lower energy state. In this case the electron has less kinetic energy and is drawn in closer to the nucleus and the atom is smaller.

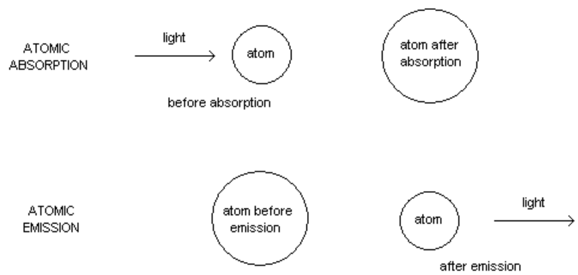


Figure 1: Atomic Absorption and Emission

The energy states atoms and molecules can assume have certain discrete values; that is, the energies of these systems are quantized. Niels Bohr determined the allowed energy states for hydrogen atoms. According to Bohr's theory of the hydrogen atom, the electron in the atom may have only the energies described by the equation

$E_n = - R_H / n^2$ (1)

where R_H is a constant with the value 2.179 x 10^{-18} J and n is the principal quantum number. An electron in the lowest energy state, n = 1, has an energy

$E_1 = -2.179 \times 10^{-18} / 1^2 = -2.179 \times 10^{-18}$ J

This energy is negative, which indicates a bound state. A positive amount of energy equal to or greater than this amount would have to be added to the electron in order to remove it from the atom. The n = 1 state also represents the smallest possible size for the atom, which according to Bohr theory has a radius, r_n, given by the equation

$r_n = n^2 a_o$ (2)

where a_o is a constant known as the Bohr radius and has the value 0.0529 nm. Since the radius increases as n increases, then when an electron is promoted to a higher energy state, the atom becomes larger; that is, the electron has more kinetic energy and can move further away from the nucleus. When all the electrons in an atom are in their lowest possible energy states, the atom is in its **ground state**.

For an electron in the next energy level, n = 2, the energy is

$E_2 = -2.179 \times 10^{-18} / 2^2 = -5.448 \times 10^{-19}$ J

Whenever an electron in an atom is in an energy state other than its ground state, the atom is in an **excited state**. Table 1 summarizes the electron energies and the atomic radii for the first four energy states of a hydrogen atom according to Bohr theory.

Table 1: Energies and Radii for Hydrogen Atom

quantum number	energy	radius
n = 1	-2.179×10^{-18} J	0.0529 nm
n = 2	-5.448×10^{-19} J	0.212 nm
n = 3	-2.421×10^{-19} J	4.77 nm
n = 4	-1.362×10^{-19} J	8.47 nm

When the electron in a hydrogen atom undergoes a transition from the excited state (n = 2) to ground state (n = 1), the energy of the atom decreases by ΔE where

$\Delta E = E_{final} - E_{initial} = E_1 - E_2 = (-2.179 \times 10^{-18}\ J) - (-5.448 \times 10^{-19}\ J)$

$\Delta E = -1.634 \times 10^{-18}$ J

According to the law of conservation of energy

$\Delta E_{system} + \Delta E_{surroundings} = \Delta E_{universe} = 0$ (3)

where ΔE_{system} represents the energy change for the atom, $\Delta E_{surroundings}$ represents the energy of the emitted photon, and $\Delta E_{universe}$ is zero since the energy of the universe does not change. Since the atom is the system and the light enters the surroundings, the previous equation can be written as $\Delta E_{atom} + \Delta E_{light} = 0$

or $\Delta E_{light} = -\Delta E_{atom} = -(-1.634 \times 10^{-18}\ J) = +1.634 \times 10^{-18}$ J

Since the energy of the hydrogen atom decreases by this much, then by conservation of energy, the energy of the surroundings must increase by a corresponding amount. This energy is observed in the surroundings as $+1.634 \times 10^{-18}$ J of electromagnetic energy (or a photon of light).

Electromagnetic energy consists of oscillating electric and magnetic fields traveling through space at 2.998×10^8 m/s. Because of its oscillating nature, electromagnetic energy can be described as a wave phenomenon, and the properties of the wave can be expressed in terms of its **wavelength**, **λ** (pronounced lambda) and its **frequency**, ***v*** (pronounced nu). The wavelength and frequency are related by the equation:

$c = \lambda v$ (4)

where c is the speed of the wave (2.998×10^8 m/s). The energy of an electromagnetic wave is directly proportional to its frequency or

$E = hv$ (5)

where h is Planck's constant and has the value 6.626×10^{-34} Js. This equation can be used to determine the frequency of the emitted radiation. (In units of s^{-1} or Hz)

In a manner similar to that described above, the wavelengths of the light given off by other electron transitions can be calculated. The results of some of these calculations for an electron in a hydrogen atom are given in Table 2.

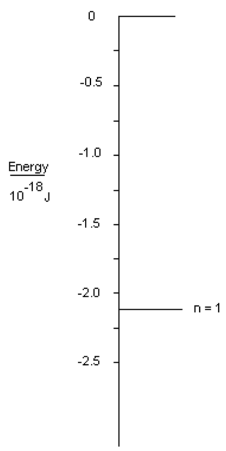
Table 2: Some Electron Transitions in a Hydrogen Atom

Electron Transition	Energy (J)	frequency (Hz)	wavelength (nm)	color
2 to 1	1.634×10^{-18}	2.466×10^{15}	121.6 nm	ultraviolet
6 to 2	4.843×10^{-19}	7.309×10^{14}	410.2	violet
5 to 2	4.576×10^{-19}	6.906×10^{14}	434.1	violet
4 to 2	4.086×10^{-19}	6.167×10^{14}	486.1	blue
3 to 2	2.960×10^{-19}	4.468×10^{14}	656.3	red

4.2: Pre-lab

The energy of each Bohr orbit with principal quantum numbers 1 through 6 is provided in the table below.

Principal Quantum Number	1	2	3	4	5	6
Energy (J)	-2.179×10^{-18}	-0.545×10^{-18}	-0.242×10^{-18}	-0.136×10^{-18}	-0.087×10^{-18}	-0.061×10^{-18}



1. Plot the energy of each orbit on the diagram above. The energy of the n=1 orbit has already been plotted.
2. Draw an arrow showing the n = 5 to n = 3 transition.
3. Calculate each of the following quantities for this transition. Show all calculations on the back of this page. Note: The energy of ANY photon is a positive number. The sign of the energy for a transition indicates whether a photon is absorbed (+) or emitted (-).

ΔE (J)	*v* (Hz)	λ (nm)

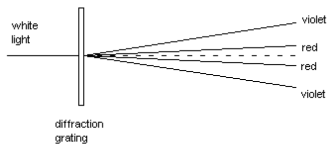
4.3: Safety

There are no hazards in this experiment.

4.4: Procedure

Part 1:

Visible light consists of all the wavelengths of electromagnetic energy from approximately 700 nm to 400 nm. White light is a mixture of all these wavelengths. White light can be separated into its component wavelengths by **refraction** through a prism or by **diffraction** using a diffraction grating.



A. Observe the spectrum of an ordinary filament light bulb through a simple tube spectroscope. This spectroscope contains a diffraction grating. Look slightly to the left or right. Do not look directly at the bulb. Carefully describe what you observe. A filament bulb produces a **continuous** spectrum since it emits all the wavelengths of visible light.

B. The various colors of light correspond to different wavelengths. To associate color with wavelength, look into the sample compartment of a Spectronic 20. This instrument contains a light bulb and a diffraction grating. The grating can be rotated to allow different colors (wavelengths) of light to pass through the sample compartment. A small angled piece of chalk has been placed in the sample compartment to reflect the light. Slowly rotate the wavelength control from 700 nm to 350 nm. Record the color of light observed at the following wavelengths.

wavelength	680 nm	590 nm	575 nm	520 nm	480 nm	410 nm
color						

C. When a high voltage is applied across a tube containing a gaseous element, the atoms absorb energy and electrons are promoted (excited) to higher energy levels. When the electrons return to lower energy levels, light is emitted. Observe the spectrum of a helium discharge tube through a spectroscope containing a diffraction grating. Do not look directly at the tube, but look slightly to the left or right. Make a sketch of what you observe (There should only be 4 lines. If you see two lines of the same color, sketch the more predominant one).

| | | | | | | | | | | | | | | | | |

700 nm 600 nm 500 nm 400 nm

D. How does the helium spectrum in part C differ from the spectrum in part A?

E. Observe the spectrum of a hydrogen discharge tube through the spectroscope. The spectroscope contains a grating and a lighted wavelength scale on which to read the wavelengths of the spectral lines being observed. Sketch the hydrogen spectrum. There are four lines in the visible portion of the hydrogen spectrum; however, the one nearest the ultraviolet is very faint, and you may not be able to see it. Draw the three lines that are visible.

| | | | | | | | | | | | | | | | | |

700 nm 600 nm 500 nm 400 nm

H. How are the hydrogen and helium spectra similar? Different?

I. Calculate the wavelength of the light emitted when an electron in a hydrogen atom undergoes a transition from n = ________ to n = _________. (The values for n will be assigned by your instructor.) Show all your calculations. Express the final answer in nanometers.

a. Find the energy for each energy level using equation (1).

b. Determine the difference in energy between the two energy levels. $\Delta E = E_{final} - E_{initial}$.

c. Determine the energy of the photon either emitted or absorbed. (The energy should have a positive value.)

c. Find the frequency using equation.

d. Find the wavelength (in nanometers) using equation.

l = ____________

Part 2:

Additional Calculations

1. Label these types of electromagnetic radiation from 1 – 4 in order of **increasing wavelength** (1 is shortest, 4 is longest).

- radio waves
- infrared radiation
- microwaves
- ultraviolet radiation

1. Calculate the **wavelength** and **energy** of each frequency of electromagnetic radiation:

• 100.2 MHz (typical frequency for FM radio broadcasting)

Wavelength = _______________

Energy = ________________

• 1070. kHz (typical frequency for AM radio broadcasting)

Wavelength = _______________

Energy = ________________

• 2.45 GHz (typical of a microwave oven)

Wavelength = _______________

Energy = ________________

CHAPTER OVERVIEW

5: Experiment 5- Chemical Structures and Modeling

5.1: Background Information

Atoms are held together by electrical forces to form molecules. The forces that hold the atoms together are electrical in nature. Consider a simple molecule like H_2. This molecule consists of two atoms of hydrogen. What holds the two atoms together to make a molecule? Each hydrogen atom has a positively charged nucleus. As you know from basic physical science, like electric charges repel; therefore, these two positive charges repel each other and try to push the two atoms apart. However, if the two electrons (one from each hydrogen atom) are in the region between the two nuclei, the atoms are held together by the attractive forces exerted on the two nuclei by the negatively charged electrons. The electrons are responsible for the bonding of the two hydrogen atoms. In fact, electrons are responsible for all chemical bonding.

A simple picture representing the H_2 molecule is shown below.

H **:** H

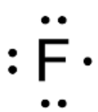
The two electrons are shown as dots. This method of representation of a molecule is often referred to as a Lewis electron dot structure. The two electrons represent an electron pair chemical bond. Electrons tend to pair up when forming bonds between atoms. The pair of electrons holding the atoms together (the chemical bond) is represented by a line as shown below.

H – H

When a single pair of electrons bond two atoms together, the bond is known as a single bond. Sometimes more than one pair of electrons is involved in bonding two atoms. Two electron pairs are known as a double bond, and three electron pairs are known as a triple bond.

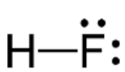
Hydrogen atoms normally form only one bond. Other atoms also have restrictions on the number of bonds they can form. As another example, consider fluorine. Fluorine has nine electrons, but not all of these electrons are involved in forming chemical bonds. The electron configuration for fluorine is $1s^22s^22p^5$. The seven outer electrons (the two 2s and the five 2p) are known as the valence electrons. Valence electrons are the ones that may be involved in chemical bonding. The two 1s electrons are known as the core electrons. The core electrons are very close to the nucleus and are not involved in chemical bonding. The Lewis dot structure for fluorine is shown below.

Lewis structures show only the valence electrons of an atom; therefore, there are seven dots (electrons).



The unpaired electron on fluorine can be shared with the electron from a hydrogen atom to form an electron pair bond to form the HF molecule. There are three pairs of electrons around the fluorine atom that are not involved in bonding. These are known as nonbonding pairs or lone pairs. By sharing an electron with hydrogen, the n = 2 electron shell of fluorine is filled and contains eight electrons. Eight electrons are known as an octet. In forming bonds with other atoms, the atoms of the second period elements (carbon, nitrogen, oxygen, and fluorine) form octets. This is also generally true for many other atoms and is known as the octet rule. (Hydrogen atoms are obviously an exception since only two electrons are needed to fill the 1s orbital.) The Lewis structure for HF can be drawn using a line to represent the bonding pair of electrons.

In counting electrons, the bonding electrons are counted for each of the bonded atoms, and the lone pair electrons are counted only for the atom on which they reside. So for HF, the hydrogen has two electrons and the fluorine has eight (an octet).



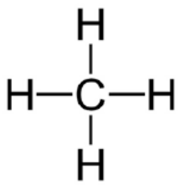
Typical Electron Arrangements Around Some Second Period Atoms in Molecules or Ions

atom	carbon	nitrogen	oxygen	fluorine
number of covalent bonds	4	3	2	1

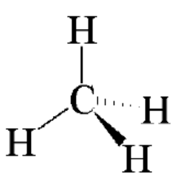
number of nonbonding pairs	0	1	2	3
total number of bonding and nonbonding electrons	8	8	8	8

(*There are exceptions for these arrangements, especially for nitrogen and oxygen*.)

Methane, CH_4, one of the simplest organic compounds, contains only hydrogen and carbon and is known as a hydrocarbon. Hydrocarbons that contain only single bonds are referred to as alkanes. The Lewis structure for methane is shown below. Note that the carbon has four bonds (an octet of electrons), and each hydrogen atom has only one bond.



A problem arises trying to represent what molecules look like by making drawings on paper. Paper is two-dimensional; molecules are three-dimensional. In the structural formula for methane shown above, it looks like the bond angles are 90^o. This is not correct. The angles are actually about 109.5^o. One scheme for representing three-dimensional structures is shown below using methane as an example.



The two solid lines represent bonds that are in the plane of the paper. The wedged line represents a bond that comes out of the plane of the paper. The dashed line represents a bond that goes into the plane of the paper. This drawing gives a three-dimensional perspective of methane. This method of representation is known as the wedge and dash method. The arrangement of the electron domains around an atom depends on the type of bonding exhibited by that atom.

Lewis Structure Exercises: A procedure for drawing Lewis dot structures for molecules of second row elements is outlined below.

1. Count the number of valence electrons in the molecule or ion. For anions, add the appropriate number of electrons. For cations, subtract the appropriate number of electrons.

2. Determine the possible connectivity of the atoms and draw a skeleton structure showing bonds (a dash represents an electron pair bond) between atoms that are bonded. It is usually a good strategy to put hydrogen atoms in the structure last so that each atom has the appropriate number of bonds.

3. Count the number of valence electrons on each atom, then distribute the remaining valence electrons so that every second-period atom has an octet. If there are not enough electrons to give them complete octets, then there must be multiple bonds between some atoms.

4. If multiple bonds are needed, repeat steps 2 and 3 with the appropriate number of multiple bonds.

5. If there is more than one possible Lewis structure, then calculate the formal charges for the atoms in each Lewis structure. The most likely structure is the one for which the atoms have the smallest formal charges and the charge distribution is consistent with the electronegativities of the atoms (negative charges reside on the more electronegative atoms).

6. Finally, check the structure to be sure the octet rule is satisfied and that the structure has the correct total number of valence electrons.

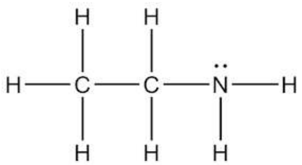
Example: Draw the Lewis structure for aminomethane, CH_5N.

1. There are a total of 14 valence electrons (4 for carbon, 1 for each of the 5 hydrogens, and 5 for nitrogen).

2. To determine the proper connectivity of the atoms, begin by bonding the carbon and the nitrogen. Next bond three hydrogens to

carbon to give the four bonds for carbon and bond two hydrogens to the nitrogen so that it has the three bonds.

3. The skeleton structure uses 12 valence electrons. Since there are a total of 14 valence electrons, two more must be added. The final two valence electrons are placed on the nitrogen as a nonbonding pair. This will give the nitrogen an octet.



4. Since all atoms have the appropriate number of electrons (octets for carbon and nitrogen and duets for hydrogen), then no multiple bonds are needed.

5. All formal charges are zero. This is the only reasonable Lewis structure.

6. This structure uses the 14 available valence electrons and the octet rule is satisfied.

5.2: Pre-lab

1. Answer the following questions for a water molecule, H_2O.

a. How many valence electrons does water have? _________

b. Draw the Lewis structure of the molecule.

c. How many electron groups are around the central atom? ________

d. How many bonding electron pairs? ________

e. How many lone pairs? ________

f. Draw the 3-D structure of the molecule.

g. What is the geometric shape of the molecule? __________________________

h. What is the bond angle between the bonding electron pairs in the molecule? ________

i. What is the hybridization around the central atom in the molecule?_______

j. Is the molecule polar or nonpolar? ___________

5.3: Safety

There are no hazards in this experiment.

5.4: Procedure

In this experiment you will construct pencil and paper models, physical models, and computer generated models of some molecules. You will use a model kit to construct the physical models. The kit consists of balls or polyhedra of various colors representing atoms. Chemists use a common color scheme to represent various types of atoms. Rigid plastic sticks are used to make bonds between atoms. The length represents the length of the bond.

Model Kit Atom Color Scheme

COLOR	ATOM	NUMBER OF BONDS
white or light blue	hydrogen	one
black	carbon	four
dark blue	nitrogen	three
red	oxygen	two
green	halogen	one
grey	transition metal	as many as six

Part 1:

1. For each of the following molecules, determine the total number of valence electrons and then draw the Lewis electron dot structure.
2. After you have drawn the correct Lewis structure for each molecule, use your model kit to make a model of each molecule. Use the curved blue rods to make multiple bonds.
3. Determine the number of electron groups, bonding groups, and lone pairs around only the central atom. Using VSEPR theory, determine the geometry, draw a 3-D structure of the molecule, and write the bond angle.
4. Using valence bond theory, determine the hybridization of the central atom of each molecule.
5. Determine if each molecule is polar or nonpolar.

LAB 5: PART 1 WILL BE HANDED OUT IN THE LAB SESSION FOR COMPLETION

Part 2:

Computational Chemistry: For this part of the experiment you, or your instructor, will use the molecular modeling software known as ***Avogadro***. Use the software to make a computer model of formaldehyde, CH_2O.

a. Open the software from the Desktop icon. Click on the **File** menu and select **New**. Click on the **Display Settings** button and check the box next to *Ball and Stick*. Make sure none of the other boxes are checked.

b. You are now ready to draw the molecule using the **Draw Tool** (pencil icon). In the drop down menu next to *Element*, select ‘Carbon (6)’, and *Bond Order*: Single. If the ‘Adjust Hydrogens’ option is checked then the appropriate number of hydrogens will be added to the molecule for normal valence. Uncheck the box to stop this from being done.

b. Move the cursor to the black, empty screen and left click. This will place one atom of carbon on the screen. Left click on the carbon atom and with the mouse button pressed, drag the mouse to a location on the right. This will create a single bond between the existing carbon atom and a new one.

c. To change the new carbon atom to an oxygen atom, select *Element*: ‘Oxygen (8)’ from the drop down menu and click on the carbon atom. It should change to a red oxygen atom.

d. To change the single bond between carbon and oxygen to a double bond, left click on the single bond. Clicking on bonds adjusts their order (single, double, triple, back to single). Right click on a particular atom (or bond) if you wish to delete it.

e. To add hydrogens to the molecule, go to the **Build** menu and select ‘Add Hydrogens’.

f. In the final step, you will optimize the molecular geometry using the **Auto Optimization Tool** (icon looks like **E** with a downward pointing arrow). Click on the icon, select *Force Field*: ‘MMFF94’ and check <u>both</u> boxes for atoms to be moveable. Click the **Start** button to run the optimization. The molecule should now have the structure

g. Click on the **Display Settings** button and view different models by selecting Wireframe, then Ball and Stick, and finally van der Waals Spheres. The space filling spheres model is most like what the molecule actually looks like. Return to Ball and Stick view at the end.

h. You will now measure bond distances and bond angles in formaldehyde using the icon that looks like a ruler (to the right of the Auto Optimization Tool), the **Measure** tool. This tool allows you to measure distance or angles. Click on two atoms to measure a distance. Click on three atoms to measure the angle between them. Right click to clear the measurement and the list of selected atoms.

i. Click on the carbon atom until a **#1** label appears on the atom. Then click on the oxygen atom until a **#2** label appears. The bond distance will be displayed at the bottom left of the screen. Record the value in the table. Clicking the right mouse button clears the labels. Determine the C – H bond distance also.

j. To measure an angle, you will need to select three atoms to define the angle. Measure the H – C = O bond angle. The result will appear at the bottom left of the screen. Also measure the H – C – H bond angle. Record the results.

C = O bond distance	C – H bond distance	H – C = O bond angle	H – C – H bond angle

k. Click on File, and then click on Close. Do not save the formaldehyde model.

l. Build a model of ammonia, NH_3, and measure the H – N – H bond angle. __________

m. Are the H – C – H (for CH_2O) **and** H – N – H (for NH_3) bond angles for the computer models consistent with the ideal bond angles? **Explain** why or why not.

CHAPTER OVERVIEW

6: Experiment 6- Intermolecular Forces

6.1: Background

In this lab, we will examine the attractive forces that hold molecules together and the disruptive forces that break them apart.

Intermolecular Forces

The forces between molecules that hold molecules together are called **Intermolecular Forces** (IMF). All material is held together by attractive forces, but there is always some disruptive force present that can break it apart. When an object is a solid at a given temperature, it means that the attractive forces must be greater than the disruptive forces. When something is a gas at any temperature it means that the disruptive forces much be much greater than the attractive forces. Finally, when the forces of disruption and attraction are on about the same level the substance will exist as a liquid.

There are only two ways that elements are held together: sharing of electrons (as in covalent bonding) or by plus to minus charge or partial charge interactions (as in ionic compounds or intermolecular forces). By contrast, there are multiple IMFs including **Dispersion forces**, **dipole-dipole forces**, **Hydrogen bonding** (H-bonds), and **Ion-Dipole forces**. In general, intermolecular forces are much weaker than in the ionic and covalent bonds that hold together the atoms and ions in a compound

Dispersion Forces

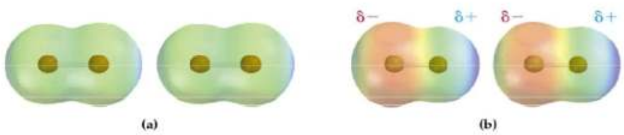


Figure 1: Dispersion forces for two Br_2 molecules (a) Represents the two molecules before the induced dipoles appear; (b) shows the alignment of the induced dipoles.

An example of dispersion forces can be seen in Br_2 (Figure 1). The temporary shift of electrons (Figure 1b) causes a temporary dipole in one molecule that will then cause (or induce) a temporary dipole in another molecule. The temporary dipoles cause quick and very weak attractions between atoms or molecules. Dispersion forces exist in all compounds and will be stronger in larger molecules or atoms that have larger numbers of electrons to shift. Larger atoms or molecules are thus more polarizable (i.e. can experience a stronger temporary dipole). Dispersion forces are the only type of intermolecular forces experienced by nonpolar molecules and single atoms.

Dipole-Dipole Forces

Dipole-dipole forces exist between **polar molecules**. Polar molecules have a permanent dipole, so the charges are separated from one another. In polar compounds, the positive end of one molecule is attracted to the negative end of another molecule (see Figure 2 for an example). Because polar molecules have a permanent dipole, rather than a temporary dipole, the dipole-dipole forces they experience are generally stronger than dispersion forces.

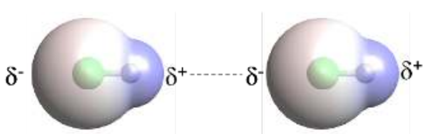


Figure 2. Dipole-dipole intermolecular interaction between two HCl molecules.

Hydrogen Bonds

Hydrogen bonds are stronger dipole-dipole forces between the hydrogen atom bonded to a N, F, or O atom to a N, F, or O in another molecule. Only molecules with a H-N, H-O, or H-F bond in them can form hydrogen bridges. These bonds are very polar; the H has a large partial positive charge, which is attracted to the lone pairs on a N, F, or O of a second molecule. Thus, the two molecules orient so that the H from the first is near the N, O, or F of the second. Two ammonia (NH_3) molecules are depicted in Figure 3 below with the Hydrogen bridge shown as a dotted lone between the H atom in one molecule and the N atom in the second molecule. Remember, the covalent bond between N and H within an ammonia molecule is considerably stronger than the hydrogen bridges holding a sample of ammonia molecules together.

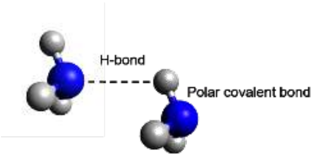


Figure 3. Hydrogen bond (dotted line) between two ammonia molecules.

Ion-Dipole Forces

Ion-dipole forces occur between an ionic substance and a polar liquid, typically a solvent. When table salt, NaCl(*s*), dissolves in water, the positive end of the water molecule is attracted to the negatively charged chloride ion (Cl^-) and the negative end of the water molecules is attracted to the positively charged sodium ion (Na^+). These attractions are strong enough to separate some ionic compounds into individual ions into solution.

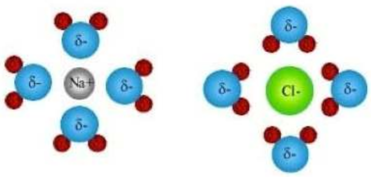


Figure 4. Ion-dipole forces showing polar water molecules orienting themselves around (left) a positively charged Na^+ ion and (right) a negatively charged Cl^- ion.

Analyzing Intermolecular Forces

The importance of identifying Intermolecular Forces is that IMFs define many physical macroscale properties of compounds and elements. The strength of an IMF determines the phase of matter of a substance, the cooling properties of a substance, the solubility of a substance, the response of a substance to electricity and the viscosity of the substance to name a few characteristics. Below, we will explore a subset of these properties and how IMFs come into play.

Viscosity

Viscosity is defined as the resistance of a liquid to flow and can be thought of as a form of inertia. Thus, a liquid with a higher viscosity will tend to be thicker and stickier, like molasses, while a liquid with a lower viscosity will flow quickly, like water. The stronger the IMFs, the more likely something is viscous.

Phase changes (melting or boiling)

Part of this experiment focuses on the Intermolecular Forces that are broken when substances undergo phase (or physical) changes. During these processes, the **chemical structure** of a substance doesn't change, only its phase changes. In a chemical equation, we can write it as $H_2O_{(l)}$ $H_2O_{(g)}$. Notice that we wrote the formula the same on both sides of the equation, so the chemical structure is the same: no **bonds** were broken during this process. We define the melting point as the temperature at which a substance is in equilibrium between solid and liquid and the boiling point as the temperature at which a substance is in equilibrium between liquid and gas. Depending on the direction of the phase change, it could either take energy (such as is needed to melt a solid) or release energy (such as released when freezing a liquid). In a solid, molecules are packed tightly together with minimal room for movement, in a liquid they can move around each other but remain close, and in a gas they are as far apart from each other as possible. Thus, the stronger the intermolecular forces, the more likely something is to be a solid.

Evaporation

When you step out of a swimming pool, you feel cool as the water evaporates off your skin. Similarly, when a metal object (e.g., a metal temperature probe) is immersed in a volatile liquid (a liquid that evaporates easily like an alcohol) and is then exposed to the air, the surface of the meal object becomes colder as the liquid evaporates. This effect is called **evaporative cooling**. Both the alcohol and the metal object are at room temperature when the object is immersed in the alcohol. In general, liquid molecules with stronger intermolecular attractions require more energy to escape to the gas phase, so they undergo a **slower** rate of evaporation. **In other words, a substance with stronger intermolecular forces will evaporate more slowly and will have more mass left after a set amount of time**.

6.2: Pre-lab

1. Which of the following substances experience **dispersion forces**? Select all that apply.
   A. arsenic trihydride (AsH_3)
   B. boron tribromide (BBr_3)
   C. hydrogen selenide (H_2Se)
   D. orthosilicic acid (H_4SiO_4)
   E. none of the above
2. Which of the following substances experience **dipole-dipole interactions**? Select all that apply.

A. antimony trifluoride (SbF_3)
B. bromine (Br_2)
C. perchloric acid ($HClO_4$)
D. silicon tetraiodide (SiI_4)
E. none of the above

3. Which of the following substances experience **hydrogen bonding**? Select all that apply.
A. carbon tetrafluoride (CF_4)
B. difluoroamine (F_2HN)
C. disulfur monoxide (S_2O)
D. sulfur tetrachloride (SCl_4)
E. none of the above

6. What **intermolecular forces** are experienced by:

A. **carbonyl selenide (COSe)?**

B. **tellurium hexafluoride (TeF_6)**?

C. **thionyl chloride ($SOCl_2$)**

D. **phosphorus triiodide (PI_3)**?

7. Rank the **boiling points** of **$AsBr_3$**, **Cl_2O** and **POF_3** in **descending** order.

8. Rank the **melting points** of **HNCO**, **N_2H_4** and **NO** in **decreasing** order.

6.3: Safety

6.4: Procedure

Coming Soon!

CHAPTER OVERVIEW

7: Experiment 7- Stoichiometry and the Chemistry of Carbonates

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7.1: Background Information

In this experiment, the goal is to identify your unknown, which is either a bicarbonate or carbonate. You will be given a two gram sample of one of the following compounds:

Na_2CO_3 K_2CO_3 $NaHCO_3$ $KHCO_3$

Bicarbonates and carbonates undergo different reactions, and they have different reaction stoichiometry. You must use the principles discussed below to correctly identify your unknown.

Decomposition

Compounds that contain the HCO_3^- ion are called bicarbonates or hydrogen carbonates. One example is sodium bicarbonate, $NaHCO_3$, which is commonly called baking soda. Hydrogen carbonates are thermally unstable and will decompose when heated according to the following chemical equation.

$2MHCO_3(s) \rightarrow M_2CO_3(s) + H_2O(g) + CO_2(g)$ (where M is a monatomic ion such as Na^+ or K^+)

For example, if $NaHCO_3$ is heated at a high temperature the following reaction occurs.

$2NaHCO_3(s) \rightarrow Na_2CO_3(s) + H_2O(g) + CO_2(g)$

The H_2O and CO_2 escape as gases and only the solid Na_2CO_3 remains. Therefore, due to the law of conservation of matter, the mass of the solid contents at the end of the reaction should be less than the mass of the solid at the beginning.

Compounds that contain the CO_3^{2-} ion are called carbonates. Sodium carbonate, Na_2CO_3, is also known as soda ash and is an important industrial chemical used in the manufacture of glass and soaps. Carbonates are thermally stable and do not readily decompose unless heated to temperatures greater than those generally attained by Bunsen burners. Therefore, if heated only with a Bunsen burner, carbonates should not decompose, and the mass of the solid should not change over the course of the heating.

7.2: Pre-lab

Calculate the molar mass of each of the following compounds to the nearest hundredth of a gram. Record results below and in the Calculations section of the laboratory report.

$NaHCO_3$	Na_2CO_3	$KHCO_3$	K_2CO_3

Write a balanced equation for each possible chemical reaction (see the previous page for model reactions). Complete each stoichiometric calculation and write your final answer in the space provided on this sheet. Turn in this pre-lab with your lab report, and also record your answers in the Calculations section of the laboratory report.

Decomposition of $NaHCO_3$: Write balanced equation below. If no reaction occurs, write "NO REACTION."

Calculate the mass of solid Na_2CO_3 that would be formed by heating 1.000 g of solid $NaHCO_3$. __________

Calculate mass ratio of solid ending to starting material. (*Divide mass of Na_2CO_3 by 1.000 g $NaHCO_3$.*) __________

Decomposition of $KHCO_3$: Write balanced equation below. If no reaction occurs, write "NO REACTION."

Calculate the mass of solid K_2CO_3 that would be formed by heating 1.000 g of solid $KHCO_3$. __________

Calculate mass ratio of solid ending to starting material. (*Divide mass of K_2CO_3 by 1.000 g $KHCO_3$.*) __________

Decomposition of Na_2CO_3 : Write balanced equation below. If no reaction occurs, write "NO REACTION."

What mass of solid material should be left after heating 1.000 g Na_2CO_3? __________

Calculate mass ratio of solid ending to starting material. (*Divide the mass left over by 1.000 g Na_2CO_3.)* __________

Decomposition of K_2CO_3 : Write balanced equation below. If no reaction occurs, write "NO REACTION."

What mass of solid material should be left after heating 1.000 g K_2CO_3? __________

Calculate mass ratio of solid ending to starting material. (*Divide the mass left over by 1.000 g K_2CO_3.)* __________

7.3: Safety

Chemicals	Hazards	Pictograms	Required PPE
Na_2CO_3	Eye irritation		Glasses/Goggles Gloves
$NaHCO_3$	n/a	n/a	Glasses/Goggles Gloves
K_2CO_3	Skin irritation Eye irritation Respiratory irritation		Glasses/Goggles Gloves
$KHCO_3$	n/a	n/a	Glasses/Goggles

7.4: Procedure

1. Obtain a clean, dry 50 mL Erlenmeyer flask. Place the flask on a hot plate and heat the flask on high heat for 5 minutes.
2. Allow the empty flask to cool and determine its mass.
3. Using your spatula, add about half (approximately one gram) of your compound to the flask and spread it so that it just covers the bottom of the flask.
4. Determine the mass of the flask and the compound.
5. Heat the flask and contents on high heat for at least 15 minutes.
6. When the heating is complete, remove the flask from the heat and allow it to cool to room temperature and again determine its mass.
7. If your compound is a carbonate, there may be a very slight change in mass due to the evaporation of a very small amount of moisture in the sample. If your compound is a bicarbonate, there will be a significant decrease in mass due to decomposition.
8. If you think that your compound is a bicarbonate, reheat it on high heat for another 5 minutes to be sure that the decomposition is complete, allow the sample to cool, and again determine the mass.

Data

(All masses should be determined to *three decimal places*.)

Mass of flask and sample __________

Mass of empty flask __________

Mass of sample __________

Mass of flask and sample after 1st heating __________

Mass of contents (after 1st heating) __________

Mass of flask and sample after 2nd heating __________

Mass of contents (after 2nd heating) __________

Ratio __________

Calculations

Molar Masses (Units: ______________)

$NaHCO_3$	Na_2CO_3	$KHCO_3$	K_2CO_3

Theoretical Mass Ratios of Ending to Starting Material (Unitless)

Decomposition Reaction of:	$NaHCO_3$	$KHCO_3$	Na_2CO_3	K_2CO_3

1. Based on the calculated mass ratio when your sample was heated, do you think it was a carbonate or a bicarbonate? What evidence supports your conclusion?

The color of light emitted by a substance when it is heated can be an important clue to the composition of the substance. This is known as a **flame test.**

Obtain a small amount of each of your compound in small, clean, dry test tube. Add a drop of DI water to the test tube to make a concentrated solution.

Obtain a clean cotton swab and saturate the tip in the solution. Place the tip of the swab into the flame and observe the color of the flame.

Compound	NaCl	KCl	$CaCl_2$	$SrCl_2$	$BaCl_2$
Flame Color	yellow	violet	red-orange	bright red	green-yellow

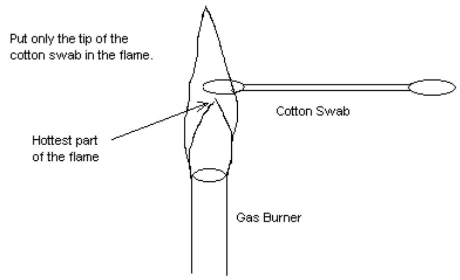


Figure: Using Cotton Swab for Flame Test

2. Using the flame test results, determine if the cation of your unknown compound is Na^+ or K^+.

What color do you observe? _________ What cation is present? ___________

3. What is the identification number for your sample? __________

Which of the four possible compounds were you given? __________

DISPOSAL: Flush all chemicals down the sink drain with plenty of water.

CHAPTER OVERVIEW

8: Experiment 8- Kinetics

8.1: Background Information

Chemical kinetics is the study of the rates and mechanisms of chemical reactions. The rate of a chemical reaction refers to the speed of the reaction. Some reactions, such as the rusting of iron, are very slow. Other reactions, such as the reaction of an acid and a base, are very fast. Chemical reactions can be classified based on their reaction kinetics (reaction order).

A rate law is a mathematical equation that describes how the speed of a chemical reaction is affected by the concentrations of its reactants. It helps determine the rate of a reaction by considering the concentration and behavior of the reactants involved.

Zero Order Reactions

Zero-order reactions (where order = 0) have a constant rate. The rate of a zero-order reaction is constant and independent of the concentration of reactants. This rate is independent of the concentration of the reactants.

The rate law is: rate = k, with k having the units of M/s

First Order Reactions

A first-order reaction (where order = 1) has a rate proportional to the concentration of one of the reactants. The rate of a first-order reaction is proportional to the concentration of one reactant.

The rate law is: rate = k[A], with k having the units of s^{-1}

Second Order Reactions

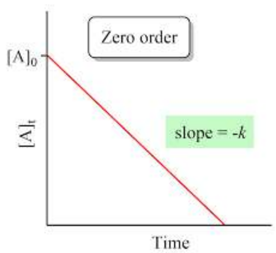
A second-order reaction (where order = 2) has a rate proportional to the concentration of the square of a single reactant or the product of the concentration of two reactants.

The rate law is: rate = $k[A]^2$, with the units of the rate constant $M^{-1}s^{-1}$

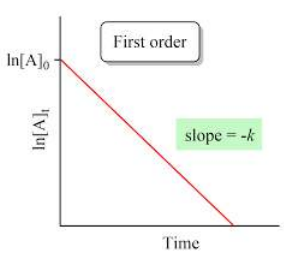
Determining Reaction Order

The order of a reaction can be determined by plotting the reaction time versus the concentrations of reactants. These are called integrated rate laws and are specific to each reaction order.

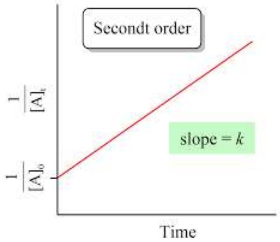
For a zero order reaction, if the time of a reaction is plotted against the concentration of the reactants, a straight line with a negative slope is seen:



For a first order reaction, when the reaction time is plotted against the natural log (ln) of the reactant concentration, a straight line with a negative slope is seen:



For a second order reaction, when the reaction time is plotted against the inverse of the reactant concentration (1/[A]), a straight line with a positive slope is seen:



For example, the thermal decomposition of NO_2 gas at elevated temperatures, occurs according to the following reaction:

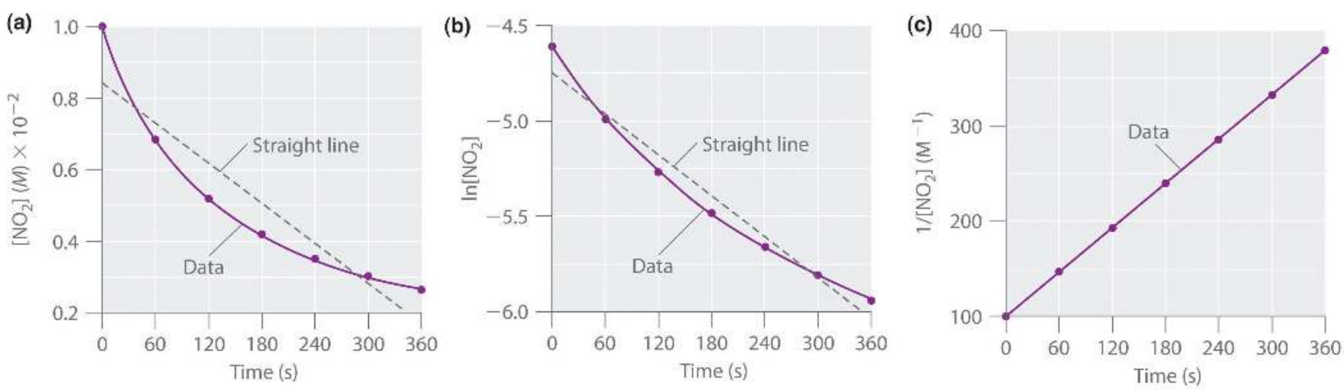
$2NO_2(g) \rightarrow 2NO(g) + O_2(g)$

Experimental data for this reaction at 330°C are listed in the table below. They are provided as $[NO_2]$, $\ln[NO_2]$, and $1/[NO_2]$ versus time to correspond to the integrated rate laws for zeroth-, first-, and second-order reactions, respectively.

Concentration of NO_2 as a Function of Time at 330°C

Time (s)	**$[NO_2]$ (M)**	**$\ln[NO_2]$**	**$1/[NO_2]$**
0	1.00×10^{-2}	−4.605	100
60	6.83×10^{-3}	−4.986	146
120	5.18×10^{-3}	−5.263	193
180	4.18×10^{-3}	−5.477	239
240	3.50×10^{-3}	−5.655	286
300	3.01×10^{-3}	−5.806	332
360	2.64×10^{-3}	−5.937	379

The actual concentrations of NO_2 are plotted versus time in part (a) shown below. Because the plot of $[NO_2]$ versus t is not a straight line, we know the reaction is not zeroth order in NO_2. A plot of $\ln[NO_2]$ versus t (part (b)) shows us that the reaction is not first order in NO_2 because a first-order reaction would give a straight line. Having eliminated zeroth-order and first-order behavior, we construct a plot of $1/[NO_2]$ versus t (part (c)). This plot is a straight line, indicating that the reaction is **second order** in NO_2.



The decomposition of NO_2: (a) the concentration of NO_2 versus t, (b) the ln[NO_2] versus t, and (c) 1/[NO_2] versus t

8.2: Pre-lab

1,3-Butadiene (CH_2=CH—CH=CH_2; C_4H_6) is a volatile and reactive organic molecule used in the production of rubber. Above room temperature, it reacts slowly to form products. Graph the data as concentration versus *t*, ln concentration versus *t*, and 1/concentration versus *t*. Then **determine the reaction order in C_4H_6 and the rate law.**

Time (s)	$[C_4H_6]$ (M)	$\ln[C_4H_6]$	$1/[C_4H_6]$ (M^{-1})
0	1.72×10^{-2}	−4.063	58.1
900	1.43×10^{-2}	−4.247	69.9
1800	1.23×10^{-2}	−4.398	81.3
3600	9.52×10^{-3}	−4.654	105
6000	7.30×10^{-3}	−4.920	137

8.3: Safety

Chemicals	Hazards	Pictograms	Required PPE
Magnesium Ribbon	Catches fire spontaneously		Glasses/Goggles Gloves
Magnesium Turnings	Catches fire spontaneously		Glasses/Goggles Gloves
0.1M, 0.25M, and 0.5M HCl	Skin irritation Eye irritation Respiratory irritation		Glasses/Goggles Gloves

8.4: Procedure

Coming Soon!

CHAPTER OVERVIEW

9: Experiment 9- Chemical Equilibrium and Le Chatelier’s Principle

9.1: Background Information

Many chemical reactions do not go to completion; that is, all of the reactants are not converted to products even though the reactants are initially present in the proper stoichiometric ratio. Consider a simple process where a molecule A is converted to 2 B molecules and as the substance B is formed some of it converts back to A. A system such as this will eventually reach a state of equilibrium and can be represented by the equation

$A \rightleftarrows 2B$

where the double arrow indicates that a state of equilibrium has been attained. The equilibrium state can be represented by the mathematical expression

$$K = [B]^2 / [A]$$

where K is the equilibrium constant. Note that the coefficients in the chemical equation are the exponents for the concentrations in the equilibrium expression.

Imagine a system where we begin with 100 molecules of A, and we monitor the amount of A and B present in the reaction system as a function of time.

time / s	0	20	40	60	80	100	120	140
[A]	100	60	40	30	25	22	22	22
[B]	0	80	120	140	150	156	156	156

Although the reactant concentrations initially decrease and the product concentrations increase, a state is finally reached where the reactant and product concentrations remain constant. It is important to recognize that this is a dynamic equilibrium; that is, the reaction has not stopped. In fact, two opposing reactions are occurring at equal rates. Reactants are being converted to products at the same rate as products are being converted back to reactants. The value of the equilibrium constant for this system can be calculated as follows:

$$K = [156]^2 / [22] = [24336] / [22] = 1106$$

Systems at equilibrium may be subjected to changes that can affect the state of equilibrium. Henri Louis Le Chatelier proposed the empirical principle that a system at equilibrium responds to a disturbance in a way that minimizes the disturbance. Explore these relationships in the pre-lab exercise.

9.2: Pre lab

1. In your own words, explain Le Chatelier's principle.

2. Coal can be used to generate hydrogen gas (a potential fuel) by the endothermic reaction:

$C_{(s)} + H2O_{(g)} \rightleftharpoons CO_{(g)} + H2_{(g)}$

If this reaction mixture is at equilibrium, predict whether each disturbance will result in the formation of additional hydrogen gas, the formation of less hydrogen gas, or have no effect on the quantity of hydrogen gas.

1. adding C to the reaction mixture
2. adding H_2O to the reaction mixture
3. raising the temperature of the reaction mixture
4. increasing the volume of the reaction mixture
5. adding a catalyst to the reaction mixture
6. adding an inert gas to the reaction mixture

9.3: Safety

Compound	Pictogram	PPE
Sodium acetate	N/A	Nitrile gloves, glasses
Potassium thiocyanate		Nitrile gloves, glasses
Iron (III) nitrate		Nitrile gloves, glasses
Methyl orange		Nitrile gloves, glasses
Ammonium hydroxide		Nitrile gloves, glasses
Ammonium chloride		Nitrile gloves, glasses
Phenolphthalein		Nitrile gloves, glasses
Cobalt (II) chloride monohydrate		Nitrile gloves, glasses
Acetic acid		Nitrile gloves, glasses
Isopropyl alcohol		Gloves, glasses

9.4: Procedure

Complex Ions:

Potassium thiocyanate, KSCN, and iron(III) nitrate, $Fe(NO_3)_3$, are ionic compounds. When solutions of these substances are mixed, the following reaction occurs. The potassium and nitrate ions are spectator ions. The product of the reaction is known as a *metal complex ion*.

$$Fe^{3+}(aq) + SCN^{-}(aq) \rightarrow Fe(SCN)^{2+}(aq)$$

Add about 1 mL of 0.001 *M* KSCN, to two adjacent wells of a 24-well plate. Place the well plate on top of a white paper towel so that you can more clearly observe any color changes. Next add 1 drop of 0.1 *M* $Fe(NO_3)_3$ solution to each of the two wells. Record your observations in the table below.

Well	#1	#2
Observations after addition of first drop of $Fe(NO_3)_3$		

What evidence do you have that a chemical reaction has occurred?

Assuming that the volume of one drop of solution is approximately 0.05 mL, complete the following table.

	mol Fe^{3+}	mol SCN^{-}	mol $Fe(SCN)^{2+}$
initial moles before any reaction			0.0 mol
moles reacted assuming reaction goes to completion			
moles after reaction assuming reaction goes to completion			

Assuming the reaction goes to completion, what is the limiting reactant and what reactant is completely consumed?

Add one additional drop of iron(III) nitrate to the solution in well #2. What do you observe? Look directly into the wells and note any small difference.

What evidence do you have that not all of the SCN^{-} was consumed in the initial reaction with Fe^{3+}? Is this consistent with the reaction achieving a state of equilibrium?

The reaction of Fe^{3+} and SCN^{-} does not go to completion, but attains a state of chemical equilibrium.

$$Fe^{3+}(aq) + SCN^{-}(aq) \rightleftharpoons Fe(SCN)^{2+}(aq)$$

Write the equilibrium constant expression for this equilibrium.

DISPOSAL: Dispose of all materials by washing down the sink with plenty of water.

Weak Acids

Acetic acid, CH_3COOH, is a weak acid. You may also see the chemical formula for this acid written as $HC_2H_3O_2$. A weak acid does not completely react with water, that is, all of the acetic acid molecules do not donate their protons to water molecules. Instead a dynamic equilibrium is attained. The equilibrium can be represented by the equation:

$$CH_3COOH(aq) + H_2O(l) \rightleftharpoons CH_3COO^{-}(aq) + H_3O^{+}(aq)$$

Write the equilibrium constant expression for this equilibrium. Equilibrium constants for weak acids are written as K_a.

Add about 2 mL of 0.10 *M* acetic acid solution to one well of a 24-well plate. Use a narrow range pH paper to determine the pH of this solution to the nearest tenth of a pH unit.

pH of 0.10 *M* acetic acid solution = _______

Using the pH, calculate the hydronium ion concentration of the solution. Show your calculations.

You will study this equilibrium by observing the color changes of an acid-base indicator known as methyl orange. Methyl orange is red or orange in solutions with relatively high concentrations of hydronium ion. At relatively low concentrations of hydronium ion, the indicator color is yellow.

Place the well plate on top of a white paper towel so that you can more clearly observe any color changes. Add a drop of methyl orange indicator to the acetic acid.

B1. What color is the resulting solution? __________

B2. Does this represent a relatively high or low H_3O^+ concentration. __________

Add a small crystal of sodium acetate, $NaC_2H_3O_2$, to the acetic acid solution. Stir with a small stirring rod until the crystal dissolves. Sodium acetate is an ionic compound and dissociates into Na^+ and $C_2H_3O_2^-$ ions when it dissolves. If no color change is observed, dissolve another small crystal of sodium acetate.

B3. What color is the resulting solution? __________

B4. Does this represent a relatively high or low H_3O^+ concentration? __________

We can understand what has happened here by applying Le Chatelier's principle. This principle states that a system at equilibrium responds to a disturbance in a way that minimizes the effect of the disturbance.

In this example, the addition of the acetate ions was the disturbance. Increasing the concentration of the acetate temporarily increased the rate of the reverse reaction to try to remove the disturbance (the additional acetate ions). In the reverse reaction, the acetate ions accepted protons from the hydronium ions to form acetic acid molecules. This reduced the concentration of the hydronium ions, which resulted in the color change of the indicator.

DISPOSAL: Dispose of all materials by washing down the sink with plenty of water.

Weak Base

The chemical equation below represents the equilibrium involved in an aqueous solution of the weak base ammonia, NH_3.

We can write a simpler representation of this equilibrium.

$NH_3(aq) + H_2O(l) \rightleftharpoons NH_4^{+}(aq) + OH^{-}(aq)$

Write the equilibrium constant expression for this equilibrium. Equilibrium constants for weak bases are written as K_b.

You will study this equilibrium by observing the color change of a chemical indicator known as phenolphthalein. This indicator is pink in solutions having a high concentration of hydroxide ions, OH^-. The indicator is colorless in solutions of low OH^- concentration.

Add about 1 mL of 0.1 *M* ammonia solution to a well of your 24-well plate. Place the well plate on top of white paper towel so that you can more clearly observe any color changes. Add a drop of phenolphthalein indicator.

C1. What color is the resulting solution? __________

C2. Does this represent a relatively high or low OH^- concentration? __________

Dissolve a small crystal of ammonium chloride, NH_4Cl, in the solution. Ammonium chloride is an ionic compound that dissociates into NH_4^+ and Cl^- ions in solution, so adding ammonium chloride increases the NH_4^+ concentration. If a color change does not occur, continue dissolving small crystals of ammonium chloride until a color change is observed.

C3. What color is the resulting solution? __________

C4. Does this indicate a relatively high or low OH^- concentration? __________

C5. What was the disturbance to the equilibrium in this case?

C6. According to Le Chatelier's principle, which direction of the reaction will be temporarily favored to bring the system back into equilibrium, the forward or the reverse? Briefly explain how this helped to minimize the disturbance.

DISPOSAL: Dispose of all materials by washing down the sink with plenty of water.

Cobalt chloride

Many equilibria can be affected by a temperature change since equilibrium constants are temperature dependent. The endothermic reaction between cobalt chloride hexahydrate, $CoCl_2{\cdot}6H_2O$, and 2-propanol, $CH_3CH(OH)CH_3$, is an example. The formula for cobalt chloride is better represented by the formula $[Co(H_2O)_6]Cl_2$ where the water molecules are directly attached to the cobalt(II) ion. In solution the water molecules are easily displaced. In a solution of alcohol, the following equilibrium occurs. The two cobalt species are different colors.

$Co(H_2O)_6^{2+}(aq) + 4Cl^-(aq) \rightleftharpoons CoCl_4^{2-}(aq) + 6H_2O(l)$

(lavender) (blue)

Dissolve about 0.25 g of cobalt chloride hexahydrate in 3 mL of 2-propanol in a 13 x 100 test tube. Use a small stirring rod to agitate the solid so that it dissolves. The solution should be deep blue. Pour half of the solution into another test tube.

CAUTION: Cobalt compounds are somewhat toxic.

D1. Use a pipet to add water dropwise to the second test tube until the solution is a lavender color. What is the predominant cobalt species after the addition of the water? Which reaction, the forward or the reverse, was favored by the addition of the water?

D2. Use a hot plate to heat a beaker of water to approximately 80 ^{o}C. Place the second tests tube in the beaker of hot water. After several minutes observe and record the color. What is the predominant cobalt species in this solution?

D3. The equilibrium was disturbed by adding heat to the system. Would you expect this disturbance to favor the forward or the reverse reaction? Remember, the forward reaction is endothermic. Is your observation in D2 consistent with this? Briefly explain.

D4. Remove the test tube from the hot water and place it in a beaker of ice water. After several minutes observe and record the color. What is the predominant cobalt species in this solution?

D5. The equilibrium was disturbed by removing heat from the system. Would you expect this disturbance to favor the forward or the reverse reaction? Is your observation in D4 consistent with this? Briefly explain.

DISPOSAL: Do **not** dispose of reagents and products from this part of the experiment by washing down the sink. Pour all cobalt chloride waste into specially marked waste containers.

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